A bounded two-tier null result for integer relations on a hybrid Khinchin-K_0 basis, with a documented capability gap and a Lean 4 statement-shape encoding

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Abstract

We test for integer relations on the hybrid basis $B_D(C) = \{1, K_0, K_0^2, \ldots, K_0^D, \log K_0\} \cup \{K_0 \cdot c : c \in C\}$ at $D = 6$ and $C = \{\pi, e, \log 2, \gamma, \zeta(2), \zeta(3), G\}$ (dimension $n = 15$). A cascading-precision PSLQ harness based on `mpmath.pslq` at $P = 2160$ dps with cascade legs at $2P$ and $4P$ returns a cascade-stable null on all 65 scanned sub-bases, independently corroborated by a PARI/GP `lindep` second leg. We report the null at two complementary tiers under an explicit verification-class taxonomy (field_standard_practice, proven_corollary): the empirical tier bounds the coefficient height $H^{\mathrm{op}}_{\mathrm{empirical}} \approx 7.997 \times 10^{69}$ at the Bailey–Borwein–Crandall (1997) confidence calibration [1], classified field_standard_practice per the Bailey–Plouffe (1998) "general rule" for PSLQ detection [2]; the rigorous tier bounds the relation Euclidean norm $H_{\mathrm{rigorous}} = 1.0361 \times 10^{72}$ via the Ferguson–Bailey–Arno (1999) Theorem 1 and Corollary 2 overshoot–free framework [3], classified proven_corollary (implying $\max_i |m_i| \geq H_{\mathrm{rigorous}}/\sqrt{n} \approx 2.676 \times 10^{71}$). Three Excluded Families delineate this bounded scope from (EF1) the BBC 1997 pure-power null, (EF2) the operator's prior signature-statistic null on π's continued fraction [5], and (EF3) Gauss–Kuzmin–Wirsing spectral-method imports. A symbolic-closure attempt in `SymPy` surveying seven candidate structural arguments (Lindemann–Weierstrass, Schanuel, Nesterenko, Mahler-measure, Galois, non-GKW CF-theoretic, height theory) plus a four-path probe returned a documented fundamental capability gap, rooted in K_0's unresolved transcendence status. The result statement is additionally encoded in Lean 4 with an explicit auxiliary-axiom trust boundary; the encoding type-checks against Mathlib4 v4.14.0 but is not a first-principles formal proof.

1 Introduction

Khinchin's constant

$$K_0 \approx 2.68545200106530644530971 48\ldots$$

is the almost-everywhere geometric mean of partial quotients of the regular continued-fraction expansion of a real number [4]. The constant is computable to arbitrary precision via classical accelerated series, but its arithmetic nature is open: no published proof of irrationality, algebraicity, or transcendence exists as of this writing. The standard expectation in the transcendence literature is that K_0 is transcendental; every published attack on irrationality has stalled at infrastructure questions rather than a clean obstruction.

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The bounded sub-question. We test, at working precision $P = 2160$ decimal places via PSLQ with a cascading-precision harness, for integer relations among the hybrid basis

$$B_D(C) \;=\; \big\{1,\, K_0,\, K_0^2, \ldots,\, K_0^D,\, \log K_0\big\} \;\cup\; \big\{K_0 \cdot c \;:\; c \in C\big\} \tag{1}$$

with

$$D \;=\; 6, \qquad C \;=\; \big\{\pi,\, e,\, \log 2,\, \gamma,\, \zeta(2),\, \zeta(3),\, G\big\}, \qquad n \;=\; \dim B_D(C) \;=\; 15, \tag{2}$$

with a coefficient bound H specified at two tiers in Section 2 below (the empirical tier as a coefficient-height bound; the rigorous tier as a Euclidean-norm bound with implied coefficient-height bound). The bounded sub-question has two terminal outcomes: a positive result (PSLQ returns a relation surviving cascade and independent verification) or a negative result (cascade-stable null with an explicit pair of height bounds and verification classes). Both terminal outcomes are first-class deliverables under the mission's explicit-evidence-acceptance-and-logging discipline.

Prior art and the named-open complement. This work sits inside a small ecosystem of prior numerical attacks on K_0's arithmetic relations. We anchor the scope on two paper-read-verified citations and one explicit methodological-cede:

(1) Bailey, Borwein, and Crandall [1] (henceforth "BBC 1997") tested at the parameters $(D{=}50, H{=}10^{70}, 7350$ dps) the pure-power basis $\{1, K_0, K_0^2, \ldots, K_0^{50}\}$ and reported a cascade-stable null (Test 1, §4). BBC's §4 close names *"log K_0, $(\log K_0)(\log 2)$, or one of many other forms involving K_0"* as open. This open passage is the named complement that the present basis $B_D(C)$ explicitly targets.

(2) The operator's prior work [5] (henceforth "the signature paper") tested, at 300 dps base with 600 dps confirmation, the signature-statistic family $\{S_n, \log S_n, S_n - K_0, \log n\}$ on π's continued-fraction partial-quotient sequence at $\mathsf{maxcoeff} = 10^3$, with 546 attempts and a 10^6 wide-cap diagnostic, returning a precision-controlled null. The Discussion section of that paper handed off, verbatim: *"Templates involving $n^{1/2}$, n^2, or algebraic combinations of K_0 with other known constants lie entirely outside the tested family and remain open."* The present basis $B_D(C)$ is precisely that "algebraic-combinations-with-other-known-constants" extension under the same AEAL discipline.

(3) The Gauss–Kuzmin–Wirsing spectral framework, while theoretically connected to K_0 via the transfer-operator point of view, is ceded methodologically: theoretical citation is permitted; method-import is prohibited per a mission-internal binding. This binding is mission-procedural; it neither weakens nor strengthens the present numerical claims.

These three prior areas form the three Excluded Families (EF1, EF2, EF3) defined formally in Section 2. The mission-relevant complement is the union

$$B_D(C) \setminus (\mathrm{EF1} \cup \mathrm{EF2} \cup \mathrm{EF3}) \tag{3}$$

i.e. the rows $\log K_0$ and $K_0 \cdot c$ for $c \in C$ together with all hybrid combinations crossing the pure-power and log/bilinear sub-blocks.

The contribution. The contribution of this manuscript is twofold and intentionally narrow:

(a) **A two-tier bounded null on the named-open complement of Eq. (3).** At the height where BBC 1997 reported a single-tier empirical null on a *different* basis (the EF1 pure-power family), this work reports a *two-tier* null on the full basis $B_D(C)$ (which *includes* the named-open EF1-complement rows): empirical $H^{\mathrm{op}}_{\mathrm{empirical}} \approx 7.997 \times 10^{69}$ (field_standard_practice, BBC-class height bound at the primary $n = 15$ sub-basis, where the operational cap is inactive) and rigorous $H_{\mathrm{rigorous}} = 1.0361 \times 10^{72}$ (proven_corollary, FBA 1999 Theorem 1 + Corollary 2; bounds the Euclidean norm of any non-trivial integer relation, which by $\|m\|_2 \le \sqrt{n}\max_i |m_i|$ implies the coefficient-height bound $\max_i |m_i| \ge H_{\text{rigorous}}/\sqrt{15} \approx 2.676 \times 10^{71}$). This is not a bound-beating result on the same basis; it is a *stronger-class* result on a basis-class BBC named open.

(b) **A documented capability gap, and an explicit Lean 4 trust boundary.** A symbolic-closure attempt for the null result surveyed seven candidate structural arguments (Lindemann–Weierstrass, Schanuel, Nesterenko, Mahler-measure, Galois, non-GKW CF-theoretic, height theory) and ran a four-path probe in `SymPy` 1.14.0. All seven candidates fail at a shared root: K_0's transcendence status is itself the major open question in this corner of the literature, and any certificate strong enough to rule out integer relations in $B_D(C)$ would imply progress on that question. We classify this as a fundamental capability gap, not a contingent machinery-availability problem. In parallel, the result statement is encoded in Lean 4 with an explicit auxiliary-axiom trust boundary; the encoding type-checks against Mathlib4, but is not a first-principles formal proof of the null. Statement-shape encoding with explicit trust boundary makes the implicit assumptions of the two-tier predicate machine-readable.

We further report, as a small methodological contribution, an operational-bound capping observation: the BBC empirical-scaling formula $H \approx 10^{P/(cn)}$ extrapolates outside the PSLQ-with-maxcoeff search range at small n, so the operational empirical bound must be reported as $\min(\text{formula}, 10^{\log_{10}\text{maxcoeff}})$ to remain testable.

Outline. Section 2 fixes the two-tier predicate, the harness architecture, the verification-class taxonomy, the PSLQ algorithm citation chain, and the Lean 4 scope statement. Section 3 reports the numerical outcomes: the 65-sub-basis empirical sweep, the primary cascade reproducibility, the PARI/GP second-leg agreement, and the two-tier bounds with H9 verification classes. Section 4 discusses the operational-bound capping observation, the M4 fundamental capability gap with its structurally-unavailable / scope-ceded sub-classification, and the basis-class distinction with respect to BBC 1997. Appendix A provides reproducibility data; Appendix B describes the mission's heuristic-accumulation methodology as a small but real meta-contribution.

2 Methods

2.1 Basis construction and three Excluded Families

The basis $B_D(C)$ in Eq. (1) is a hybrid of three sub-blocks: the pure-power tower $\{1, K_0, K_0^2, \ldots, K_0^D\}$ (dimension $D + 1 = 7$), the singleton $\{\log K_0\}$ (dimension 1), and the bilinear set $\{K_0 \cdot c : c \in C\}$ (dimension $|C| = 7$), for a total of $n = D + 1 + 1 + |C| = 15$. The Excluded Families partition the mission-relevant scope as follows.

EF1 (BBC-grandfathered pure-power subset). EF1 is the linear span of $\{1, K_0, K_0^2, \ldots, K_0^D\}$. A relation lying entirely in EF1 reproduces, at strictly weaker parameters ($D{\le}6$ vs. $D{\le}50$, $P{=}2160$ vs. 7350 dps), the BBC 1997 §4 Test 1 null. Such a finding is logged as a BBC-reproduction diagnostic rather than a primary result.

EF2 (signature-paper-grandfathered). EF2 is the signature-statistic family on π's continued fraction $\{S_n, \log S_n, S_n - K_0, \log n\}$. By construction $B_D(C)$ contains no element of EF2; the harness does not test it. The signature paper [5] is the legitimacy anchor for the present extension.

EF3 (GKW-method-ceded). EF3 covers methodological imports from the Gauss–Kuzmin–Wirsing transfer-operator spectral theory. Theoretical citation is permitted; method-import is prohibited. If PSLQ surfaces a relation whose interpretation requires GKW machinery, the harness halts and the relation routes to the mission's mutation log for operator review rather than into this manuscript.

2.2 The two-tier predicate

A run of the harness on a sub-basis $B \subseteq B_D(C)$ produces a *cascade-stable null* if `mpmath.pslq` returns `None` on B at all three precisions $\{P, 2P, 4P\} = \{2160, 4320, 8640\}$ dps with the same maxcoeff bound 10^{70} and the H9 maxsteps schedule (Section 2.5), AND a second-leg PARI/GP `lindep` run on B returns either no relation or a relation whose largest coefficient exceeds the empirical operational height $H^{\mathrm{op}}_{\mathrm{empirical}}$.

For a cascade-stable null we report at two height tiers under the H9 verification-class taxonomy (Section 2.4):

Tier 1 — empirical, classified field_standard_practice. At BBC-1997 confidence calibration, no integer relation among the basis elements has all coefficients of size $\leq H_{\mathrm{empirical}}$, where

$$H_{\mathrm{empirical}} \;=\; 10^{P/(cn)}, \qquad c \approx 2.06,\ P = 2160,\ n = 15 \;\Rightarrow\; H_{\mathrm{empirical}} \;\approx\; 7.997 \times 10^{69}. \tag{4}$$

The operational tier is the capped value $H^{\mathrm{op}}_{\mathrm{empirical}} = \min(H_{\mathrm{empirical}}, 10^{\log_{10} \mathsf{maxcoeff}}) = \min(H_{\mathrm{empirical}}, 10^{70})$; both are reported (Section 3.2). The verification class field_standard_practice reflects that the bound $H \approx 10^{P/(cn)}$ is community-validated empirical practice (the BBC c-calibration over the bare Bailey–Plouffe "general rule" of [2, §2], see Section 2.4), not a theorem of the underlying algorithm.

Tier 2 — rigorous, classified proven_corollary. At the FBA 1999 Theorem 1 + Corollary 2 conservative form, every non-trivial integer relation m among the basis elements has Euclidean norm $\|m\|_2 \geq H_{\mathrm{rigorous}}$, where H_{rigorous} is the largest value the per-iteration lower-bound certificate $M = 1/\max_{i,j} |H_{i,j}|$ attains across the cascade, scaled by mpmath's documented integer-truncation safety factor:

$$H_{\mathrm{rigorous}} \;=\; 100 \cdot \max_{k \in \mathrm{cascade}} \mathrm{reported\_norm}_k. \tag{5}$$

The convention $\|\cdot\|_2$ for the relation norm follows the verbatim Step 5 statement of [2, §2]: *"there can exist no relation vector whose Euclidean norm is less than M"*; the per-iteration certificate is the [3, Theorem 1] bound $M_x \geq 1/\max_i |h_{i,i}|$ on the smallest relation Euclidean norm. Equivalently, since $\|m\|_2 \leq \sqrt{n}\,\max_i |m_i|$, the coefficient-height bound implied by Tier 2 is $\max_i |m_i| \geq H_{\mathrm{rigorous}}/\sqrt{n}$ at the relevant sub-basis dimension. The verification class proven_corollary reflects that [3, Theorem 1] is a rigorous_theorem-class result and Eq. (5) is its conservative proven-corollary application.

2.3 Algorithm-chain citation and γ boundary case

The chain from this manuscript's bound back to the underlying theorem runs through four citations:

$$\begin{aligned} \texttt{mpmath.pslq} &\longrightarrow \text{Bailey–Plouffe 1998 §2 [2]} \\ &\longrightarrow \text{FBA 1992 preprint} \longrightarrow \text{Ferguson–Bailey–Arno 1999 [3].} \end{aligned} \tag{6}$$

`mpmath.pslq`'s docstring cites [2, §2] verbatim (URL http://www.cecm.sfu.ca/organics/papers/bailey/paper/html/node3.html; SHA-256 hashes recorded in Appendix A); [2, §2] explicitly presents PSLQ in its "simpler formulation" and cites the Ferguson–Bailey complex PSLQ preprint (circulated as a 1992 technical report) as its reference [13], the same algorithm that became [3]. The norm-bound certificate $M = 1/\max_j |H_j|$ appearing in [2, §2 pseudocode step 5] is equivalent in conservative form to the per-iteration certificate $M_x \geq 1/\max_i |h_{i,i}(k)|$ of [3, Theorem 1].

The rigorous tier of this manuscript invokes [3, Theorem 1] and [3, Corollary 2] specifically; it does *not* invoke [3, Theorem 2] or [3, Theorem 3]. Both omitted theorems carry a strict $\gamma > \sqrt{4/3}$ condition ([3, Definition 5]), and [3, §7] explicitly states that their conclusions "make no sense or have no apparent content" at $\gamma \leq \sqrt{4/3}$ in the real case; the paper does not extend either result to the boundary $\gamma = \sqrt{4/3}$ by continuity or any other argument. Theorem 1's per-iteration certificate $M_x \geq 1/\max_i |h_{i,i}(k)|$, by contrast, is established without invoking the $\gamma > \sqrt{4/3}$ condition — its proof is structurally γ-independent — so the certificate holds at $\gamma = \sqrt{4/3}$ directly. The `mpmath.pslq` implementation chooses $\gamma = \sqrt{4/3}$ exactly internally (line `g = sqrt_fixed((4«prec)//3, prec)`), which lies inside the parameter scope explicitly treated in [2, §2]. The complementary iteration-count contrapositive used in `harness/rigorous_bound.py` — the inference that if `mpmath.pslq` reaches the `maxsteps` schedule's iteration budget without declaring relation discovery, then no relation of Euclidean norm $\leq$ the terminal certificate exists — is anchored on the γ-range-independent PSLQ re-presentation of [2, §2], not on [3]'s $\gamma > \sqrt{4/3}$-conditional Theorem 2. We do not invoke Theorem 3.

2.4 The H9 verification-class taxonomy and the Bailey–Plouffe "general rule"

We report all numerical claims under a four-class verification taxonomy ("H9" in the mission's internal heuristic numbering): rigorous_theorem, proven_corollary, field_standard_practice, empirical_heuristic. The class field_standard_practice corresponds precisely to what [2, §2] terms a "general rule" for PSLQ detection. The relevant passage reads:

> *"As a general rule, one can expect to detect a relation of degree n, with coefficients of size 10^m, provided that the input vector is known to somewhat greater than $m \cdot n$ digit precision."* [2, §2]

This is the canonical statement of the $H \approx 10^{P/n}$ empirical-scaling folklore; BBC 1997 refines this to $H \approx 10^{P/(cn)}$ with a calibrated confidence factor $c \approx 2.06$ in the exponent denominator, absorbing the "somewhat greater than" hedge. The class field_standard_practice formalizes this primary-source distinction: a community-validated empirical scaling stated as a rule of practice, not as a theorem of PSLQ.

The class proven_corollary is reserved for derivations from [3, Theorem 1] or [3, Corollary 2] via mpmath's conservative `norm //= 100` truncation; these are theorem-grade up to the documented $100\times$ safety factor.

Distinguishing the predicate class from the reported statistic. The H9 verification class assigned to a sub-basis record is a predicate-level classification of the cascade-stable-null result for that sub-basis, determined by which maxsteps schedule was in force at run time: the rigorous-tier schedule $(100\,000, 80\,000, 50\,000)$ is applied only to the primary $n = 15$ cascade (one record), yielding class proven_corollary; the empirical-tier schedule $(2000, 2000, 2000)$ is applied to the 64 smaller sub-bases, yielding class field_standard_practice. The H_{rigorous} column reported alongside every record is a statistic of the [3, Theorem 1] certificate evaluated at termination; its numerical magnitude varies with the sub-basis but does not by itself upgrade the record's predicate class. The rigorous certificate $\|m\|_2 \geq H_{\text{rigorous}}$ holds for every record where $H_{\text{rigorous}} > 0$, but is reported as a proven_corollary-class predicate result only for records run under the rigorous-tier schedule.

2.5 Harness architecture

The harness (`harness/verify.py` and `harness/sweep.py`) is written against `mpmath` 1.3.0 / Python 3.12.10 and shells out to PARI/GP 2.18.1 for the second leg. Each candidate sub-basis B

runs in three phases:

1. **mpmath cascade.** $\texttt{mpmath.pslq}(B, \mathsf{tol} = 10^{-\lfloor P\cdot \texttt{tol\_decimal\_factor}\rfloor}, \mathsf{maxcoeff} = 10^{70}, \mathsf{maxsteps} = \mathrm{schedule}(n))$ (with `tol_decimal_factor` = 0.6 the code default in `harness/verify.py:run_pslq_with_capture`; see Appendix A) runs at $P = 2160$ dps, then at $2P$, then at $4P$. The cascade verdict is *stable null* if all three legs return `None`; otherwise *candidate* (further classified by Excluded-Family containment) or *unstable* (transient PSLQ false positive).
2. **Rigorous-bound extraction.** `harness/rigorous_bound.py` parses `mpmath.pslq`'s verbose-mode norm output to extract the per-iteration certificate of [3, Theorem 1], returning H_{rigorous} as in Eq. (5). The module docstring documents four divergences (D1–D4) between mpmath's internal form and the canonical [3, Theorem 1] statement; divergences D1 and D4 are conservative (mpmath's bound is weaker than the canonical form), and divergences D2 and D3 are cleared by the paper-read of [2, §2] (Section 2.3).
3. **PARI/GP second leg.** `harness/gp_lindep.py` writes the basis to a temporary file and invokes `gp -q -default parisize 4000000000` on a small script that returns the smallest `lindep` vector at the same precision. A second-leg *noise-above-bound* verdict (gp_noise_relation_above_H_empirical_confirms_null) is recorded when the smallest `lindep` relation has $\max|m_i| > H^{\text{op}}_{\text{empirical}}$, confirming the null at the empirical tier.

The cascade-stability check is non-collapsible: all three precision legs must independently return `None` for the verdict. mpmath's PSLQ implementation is deterministic at fixed precision and seed on the build used here (`mpmath 1.3.0` on Python 3.12.10); see Section 3.5.

2.6 Operational-bound capping clause

The BBC empirical-scaling formula $H^{\text{fo}}_{\text{empirical}} = 10^{P/(cn)}$ extrapolates outside the algorithm's maxcoeff search range at small n for our parameters: at $P = 2160$ dps, $c = 2.06$, the formula evaluates to $\approx 10^{524}$ at $n = 2$ and $\approx 10^{349}$ at $n = 3$, both far above the maxcoeff $= 10^{70}$ search bound. A PSLQ-with-maxcoeff run cannot detect any integer relation whose largest coefficient exceeds maxcoeff, regardless of the formula's extrapolation. We therefore report the operational empirical bound as the cap

$$H^{\text{op}}_{\text{empirical}} = \min\left(H^{\text{fo}}_{\text{empirical}},\ 10^{\log_{10}\mathsf{maxcoeff}}\right). \tag{7}$$

At the primary sub-basis $n = 15$, the formula yields $H^{\text{fo}}_{\text{empirical}} \approx 7.997 \times 10^{69} < 10^{70}$, so the cap is inactive and $H^{\text{op}}_{\text{empirical}} = H^{\text{fo}}_{\text{empirical}}$. At small n, the cap activates and $H^{\text{op}}_{\text{empirical}} = 10^{70}$. Both values are retained in the JSONL output for transparency (see Section 4).

2.7 Lean 4 statement-shape encoding

The result statement is encoded in Lean 4 with documented auxiliary axioms; the encoding type-checks against Mathlib4 v4.14.0. Full formal verification requires further development beyond this mission's scope. The Lean 4 skeleton (approximately 385 source lines across three modules `M5/Definitions.lean`, `M5/Result.lean`, `M5/Axioms.lean`) makes the result's dependency structure machine-readable: the axiom set behind `m3_null_at_rigorous_bound` is exactly

$$\{\texttt{pslq\_cascade\_implies\_no\_integer\_relation}\} \cup \{\text{nine opaque constant axioms}\}$$
$$\cup\ \{\text{Lean core: } \texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}\},$$

as recorded in `m5/_print_axioms_output.txt` (committed at `gold/M5 = 208f6fc`).

What the encoding IS and is NOT. Statement-shape Lean 4 encoding with explicit trust boundary is a methodological contribution distinct from the numerical-and-rigorous-bound K_0 result: it makes the implicit assumptions of the two-tier predicate explicit and machine-checkable. The encoding is NOT a formal proof of the null result. The M5 theorem discharges via `exact pslq_cascade_implies_no_integer_relation` — an `axiom` declaration, not a derivation. We avoid the phrasing "formally verified in Lean 4" and "Lean 4 proof of the null result" for this reason; the correct phrasing is "statement-shape machine-checkable."

Auxiliary-axiom enumeration. The trust boundary comprises four classes of axioms in `m5/M5/`. First, nine constant axioms (`realPi`, `realE`, `realLn2`, `zeta2`, `zeta3`, `eulerMascheroni`, `catalanConst`, `K0`, `logK0`) encode named real-valued constants as opaque `axiom` declarations rather than imports from Mathlib's full constructions; this is a scope simplification (Mathlib v4.14.0 provides `Real.pi`, `Real.exp`, `Real.log` but not `Real.catalan` or `Real.eulerMascheroni`). Second, two evidence axioms (`pslq_cascade_implies_no_integer_relation` and `bbc_empirical_implies_no_integer_relation`) encode the load-bearing claims as axioms, not theorems; they are justified informally by [3, Theorem 1 + Corollary 2] and BBC 1997 respectively, but are not derived in Lean. Third, one acknowledgement axiom (`M4_fundamental_gap`) is a documentary placeholder whose role is only to provide a Lean-level handle for the M4 fundamental-gap proposition (Section 4.3). Fourth, one numerical-binding stub (`K0_value_first_digits`) is documentary only and not used by the result theorem.

Mathlib cache operational note. The Mathlib4 v4.14.0 package cache available to this work was a build snapshot from a sibling project on the same machine. Lake's `Replayed` mechanism re-uses pre-built `.olean` files when their cryptographic trace-hash matches the expected hash; this is fast (sub-second per module). However, the cache contains only the import closure of the sibling project's source files, not the full Mathlib library uniformly built. An initial M5 implementation importing `Mathlib.Data.Real.Pi.Bounds` plus `Mathlib.Analysis.SpecialFunctions.{Log.Basic, Exp}` found 587 modules replayed (matching the sibling project's import set) and then began compiling modules outside that closure from scratch. The mission resolved this operational scope by simplifying M5 imports to `Mathlib.Data.Real.Basic` plus `Mathlib.Data.List.Basic` (both inside the cache closure) and encoding the named constants as opaque axioms as above. The choice of opaque axioms for these constants is therefore not a defect of the formalization but an operationally honest adaptation to the available cache closure, consistent with the statement-shape scope.

3 Results

3.1 Aggregate verdict on the 65-sub-basis sweep

The canonical sweep `harness/sweep_output/m32b_empirical_sweep.jsonl` (frozen at `gold/M3` = `9c2702d`) records 65/65 sub-basis verdicts of cascade-stable null, with the per-tier verification classes summarized in Table 1.

No anomalies surfaced. No halt conditions triggered.

3.2 The primary cascade and its two-tier bounds

The primary sub-basis is the full $B_D(C)$ at $n = 15$ with `maxcoeff`$=10^{70}$ and the cascade precision schedule $\{P, 2P, 4P\} = \{2160, 4320, 8640\}$ dps with maxsteps schedule $(100\,000, 80\,000, 50\,000)$. The reproducibility check across two independent executions (M3.2a at commit `d55ffbc`; M3.2b at commit `f621641`) returns bit-for-bit identical outputs (Table 2).

Table 1: Aggregate verdict across the 65-sub-basis sweep.

Quantity	Threshold	Measured	Status
Cascade-stable null verdicts	65/65	65/65	PASS
`primary_full` reproducibility, M3.2a↔M3.2b	K, H_{rigorous} identical at $P, 2P, 4P$	identical to 71 digits	PASS
`primary_full` verification class	proven_corollary	proven_corollary	PASS
64 empirical sub-bases verification class	field_standard_practice	64/64 field_standard_practice	PASS
PARI/GP second-leg per candidate	noise relation above $H^{\text{op}}_{\text{empirical}}$	65/65	PASS
H10 pre-canonical full-regime dry-run	clean across all 65	clean (113 s)	PASS

Table 2: Primary cascade outcomes, identical between M3.2a and M3.2b at every cascade leg (mpmath 1.3.0 / Python 3.12.10).

P (dps)	K iterations	reported_norm (71-digit int)	H_{rigorous}
2160	29 363	$\underbrace{10360617603\ldots338548597674}_{71\text{ digits}}$	1.0361×10^{72}
4320	29 363	(identical)	1.0361×10^{72}
8640	29 363	(identical)	1.0361×10^{72}

The two-tier bounds for the primary cascade are

$$\begin{aligned}
H^{\text{fo}}_{\text{empirical}} &= 10^{P/(cn)} \approx 10^{2160/(2.06\cdot 15)} \approx 7.997 \times 10^{69} && \text{(class field_standard_practice)},\\
H^{\text{op}}_{\text{empirical}} &= \min(H^{\text{fo}}_{\text{empirical}}, 10^{70}) = 7.997 \times 10^{69} = H^{\text{fo}}_{\text{empirical}} && \text{(cap inactive at } n = 15),\\
H_{\text{rigorous}} &= 1.0361 \times 10^{72} && \text{(class proven_corollary; Eq. (5))}.
\end{aligned}$$

In words: no integer relation among the basis elements of $B_D(C)$ has all coefficients $\leq 7.997 \times 10^{69}$ at the BBC-1997 empirical confidence calibration of [2, §2]; and, by [3, Theorem 1 + Corollary 2] applied to the cascade's terminal norm-bound certificate, every non-trivial integer relation has Euclidean norm $\geq 1.0361 \times 10^{72}$ (proven_corollary-class), implying the coefficient-height bound $\max_i |m_i| \geq 1.0361 \times 10^{72}/\sqrt{15} \approx 2.676 \times 10^{71}$.

3.3 Per-family aggregate

The 65 sub-bases group into six families (Table 3). The primary cascade ($n = 15$) dominates the total wall-clock at 90.5%; the remaining 64 empirical candidates aggregate to 283.9 s ($\sim$ 4.7 min). Small-n families ($n = 2, 3$) terminate fast with high cancellation-norm, yielding H_{rigorous} statistic values in the 10^{72}–10^{73} range; mid-n families ($n = 8, 9$) run under the empirical-tier maxsteps schedule and yield H_{rigorous} statistic values materially below $H^{\text{op}}_{\text{empirical}}$. For all 64 empirical sub-bases the predicate verification class is field_standard_practice (set by the maxsteps schedule, per § 2.4), independent of the reported H_{rigorous} statistic; only the primary $n = 15$ sub-basis carries the proven_corollary class.

The wall-clock total (49.6 min) is approximately 30% under our 70–90 min pre-execution estimate, dominated by the primary $n = 15$ cascade.

3.4 PARI/GP independent corroboration

For each of the 65 sub-bases, the harness runs an independent `gp lindep` second leg on the same basis at the same precision. On every sub-basis, `gp lindep` returns either no relation or a relation whose largest coefficient strictly exceeds the empirical operational bound $H^{\text{op}}_{\text{empirical}}$; the cascade-stable null is corroborated independently. This addresses the concern that `mpmath.pslq`

Table 3: Per-family aggregate over the 65-sub-basis sweep. The H_{rigorous} statistic is reported per record (Theorem 1 certificate at termination); the verification class follows the maxsteps schedule (§ 2.4).

Family	n	count	Σ wall-clock	H_{rigorous} median	verification class
`primary_full`	15	1	2693.8 s	1.04×10^{72}	proven_corollary
`complement_plus_K0_k`	9	7	229.8 s	1.26×10^{16}	field_standard_practice
`full_complement`	8	1	27.0 s	1.16×10^{21}	field_standard_practice
`triplet_log_*`	3	21	21.4 s	1.17×10^{72}	field_standard_practice
`pair_bilinear_*`	2	21	3.5 s	2.38×10^{72}	field_standard_practice
`pair_log_*`	2	14	2.3 s	2.34×10^{72}	field_standard_practice
Total	—	65	2977.8 s (49.6 min)	—	—

and `gp lindep`, while implementing closely related algorithms (PSLQ in `mpmath`; LLL-based `lindep` in PARI/GP), might both miss a relation through a shared algorithmic pathology; on the present basis this concern does not materialize.

3.5 Reproducibility of the cascade

The reproducibility check M3.2a $\leftrightarrow$ M3.2b returns bit-for-bit identical outputs at every cascade leg: $K = 29\,363$ iterations at all three precisions, identical 71-digit `reported_norm` value, identical H_{rigorous}. This empirically confirms determinism of `mpmath.pslq` at $n = 15$ across a $4\times$ precision range and two independent executions on the build used here. The determinism check was operator-mandated post-U-MISSION-N: any non-reproducibility would have indicated non-determinism in the algorithm that would affect the verification-class assignment.

4 Discussion

4.1 Basis-class distinction: apples-to-stronger-apples with BBC 1997

We *do not* claim a bound improvement on the BBC 1997 basis. BBC 1997 §4 Test 1 tested EF1 (the pure-power family $\{1, K_0, \ldots, K_0^{50}\}$ at $D = 50$) at $H \leq 10^{70}$ and 7350 dps; the present work tests the full basis $B_D(C)$ ($D = 6$, $|C| = 7$, $n = 15$, which contains both EF1 at $D{\leq}6$ and the named-open EF1-complement rows $\{\log K_0\} \cup \{K_0 \cdot c\}$) at $P = 2160$ dps. The bases differ in two structural respects — EF1 truncation and EF1-complement inclusion — so a direct "we improve BBC's 10^{70} to 10^{72}" framing would be apples-to-apples on bases that are not the same and is therefore avoided. The honest comparison is:

(i) BBC 1997 produced a single-tier empirical null on EF1 at $H = 10^{70}$, sufficient to rule out a low-degree polynomial relation with bounded height on the pure-power basis.

(ii) The present work produces a two-tier null on $B_D(C)$ (which includes EF1 at $D{\leq}6$ and adds the EF1-complement): empirical $H^{\text{op}}_{\text{empirical}} \approx 7.997 \times 10^{69}$ on coefficient height (field_standard_practice, BBC-class at the primary $n = 15$ sub-basis) and rigorous Euclidean-norm bound $H_{\text{rigorous}} = 1.0361 \times 10^{72}$ (proven_corollary, FBA-class; implied height $\max_i |m_i| \geq 2.676 \times 10^{71}$). The rigorous tier is the methodological strengthening; the EF1-complement inclusion is the named-open basis-class extension.

A future apples-to-apples comparison would test a union basis that preserves BBC's $D = 50$ pure-power tower and adds the EF1-complement, giving dimension $n = (D{+}1){+}1{+}|C| = 51{+}1{+}7 = 59$. The wall-clock cost scales roughly as $\sim 59^3/15^3 \approx 61\times$ per cascade step, which is feasible on an extended budget but is outside the present scope.

4.2 Operational-bound capping observation

Across the 64 empirical sub-bases of `m32b_empirical_sweep.jsonl`, the canonical operational empirical bound $H^{\mathrm{op}}_{\mathrm{empirical}} = \min(10^{P/(cn)},\ 10^{\log_{10}\mathsf{maxcoeff}})$ differs from the uncapped BBC scaling formula $H^{\mathrm{fo}}_{\mathrm{empirical}} = 10^{P/(cn)}$ at every $n \leq 14$ sub-basis: at $P = 2160$ dps, $c \approx 2.06$, $\mathsf{maxcoeff} = 10^{70}$, the formula yields $\approx 10^{70}$ exactly at $n \approx 14.97 \approx 15$. At $n = 15$ the cap is inactive ($H^{\mathrm{op}}_{\mathrm{empirical}} = H^{\mathrm{fo}}_{\mathrm{empirical}} \approx 7.997 \times 10^{69}$). At small n ($n = 2, 3$), the formula evaluates well above the maxcoeff bound ($\sim 10^{524}$ and $\sim 10^{349}$ respectively); reporting these as operational bounds would overclaim the searched range. We retain both values in the JSONL output: $H^{\mathrm{op}}_{\mathrm{empirical}}$ for predicate evaluation and table reporting, $H^{\mathrm{fo}}_{\mathrm{empirical}}$ for transparency about where the empirical extrapolation diverges from the searched range.

This is a small refinement to BBC-class empirical-tier reporting at parameter regimes where the scaling formula extrapolates outside the algorithm's maxcoeff bound. The refinement is grounded in the "somewhat greater than" hedge of [2, §2]: the empirical scaling is stated as a rule of practice, not as a tight bound; reporting it as if it were a tight bound at parameter regimes outside the search range is the failure mode this capping clause addresses.

4.3 Capability gap: symbolic closure of the null result is fundamental

Symbolic closure of the M3 null result was attempted via a survey of seven candidate structural arguments (Lindemann–Weierstrass, Schanuel, Nesterenko, Mahler-measure, Galois, non-GKW continued-fraction-theoretic, and height theory) and a four-path probe in `SymPy` 1.14.0 (`is_transcendental`/`is_irrational`, `nsimplify` against a named-constants basis, Groebner basis on the symbolic basis $B_D(C)$, and transcendence-theorem applicability check). Every candidate failed at the same root: K_0's transcendence status is itself the major open question in this corner of the literature, and any structural certificate strong enough to rule out integer relations in $B_D(C)$ at relation Euclidean norm $\leq 1.0361 \times 10^{72}$ would imply progress on that question. We classify this gap as fundamental: no known structural result is strong enough — not merely a machinery-availability problem — and the null result therefore stands on numerical and rigorous-bound evidence per the two-tier predicate ($H^{\mathrm{op}}_{\mathrm{empirical}} \approx 7.997 \times 10^{69}$ field_standard_practice; $H_{\mathrm{rigorous}} = 1.0361 \times 10^{72}$ proven_corollary on the Euclidean norm).

Sub-classification of the failure modes. The seven candidates fail for two qualitatively different reasons. Six of seven are structurally unavailable on external grounds alone: Lindemann–Weierstrass requires $K_0 = e^{\alpha}$ for algebraic α, which is not known to hold; Schanuel is unproven, and a clean $K_0 = \exp(z)$ representation for explicit $\mathbb{Q}$-linear z is not known; Nesterenko's 1996 method covers $\pi, e^{\pi}, \Gamma(1/4)$ and has no published extension to the Khinchin class; Mahler measure, Galois theory, and height theory all require K_0 algebraic with known minimal polynomial, which is unknown. The seventh (non-GKW CF-theoretic) is internally scope-ceded under a mission-procedural binding, but is also structurally unavailable externally: no non-GKW continued-fraction-theoretic result strong enough to constrain integer-relation existence in $B_D(C)$ is published. The fundamental classification is therefore based primarily on the structural unavailability of the six external candidates, with the seventh as a double-blocked corroborating case.

What we are not claiming. We are not claiming structural non-existence of integer relations on $B_D(C)$, nor a structural closure of the M3 null. The null result stands on its numerical evidence and rigorous-bound proven-corollary application alone; the symbolic-closure attempt is reported here for transparency about the limits of what numerical evidence plus available structural theory can jointly establish for this constant class.

A Reproducibility

Repository and gold tags. All canonical artifacts are hosted at https://github.com/papanokechi/khinchin-k0-bounded on `origin/main`, tagged in milestone order: `gold/M1` (`15b216f`; target selection), `gold/M2` (`ca9c989`; M2.3 predicate), `gold/M3` (`9c2702d`; canonical JSONL outputs), `gold/M4` (`fd13eeb`; capability-gap document), `gold/M5` (`208f6fc`; Lean 4 skeleton). Tags are annotated; the manuscript snapshot is at the commit cited in the title-page footnote.

Environment. The canonical environment lock is in `env/M1.lock` (generated at the M1.3 gold freeze): Microsoft Windows 11 Pro Insider Preview 10.0.29585; Python 3.12.10; `mpmath` 1.3.0; `sympy` 1.14.0; `numpy` 2.4.4; `gmpy2` 2.3.0; Lean 4.29.1; Lake 5.0.0-src+f72c35b; Mathlib4 v4.14.0; PARI/GP 2.18.1.

Commands.

```
# M3.2a primary cascade only (n=15 full basis):
python harness/sweep.py --m32-primary-measurement --m32-primary-greenlighted

# M3.2b empirical sweep (65 sub-bases, ~50 min on a 2024-class CPU):
python harness/sweep.py --full --m32-full-greenlighted

# H10 pre-canonical full-regime dry-run (113 s wall-clock):
python harness/sweep.py --m31-extended-dry-run

# Lean 4 statement-shape check (post 'lake update'):
cd m5 && lake build M5
cd m5 && lake env lean PrintAxioms.lean # reproduces _print_axioms_output.txt
```

Tolerance parameter (code-default). The canonical run uses `tol_decimal_factor` $= 0.6$ as a code default in `harness/verify.py:run_pslq_with_capture`; this factor sets `mpmath.pslq`'s `tol` argument to $10^{-\lfloor P\cdot 0.6\rfloor}$ at working precision P. The value is fixed in source code rather than passed at invocation, so reproducers using the canonical commands above inherit `tol_decimal_factor` $= 0.6$ automatically; any divergence requires a source-level patch with a fresh canonical-artifact regeneration.

Paper-read primary sources (SHA-256 hashes). The following primary sources were paper-read and their digital copies hashed at retrieval:

Source	lit-NNN	SHA-256 (file)
Signature paper [5]	lit-001	(operator-authored, MIT-licensed repo snapshot)
BBC 1997 [1]	lit-002	(paper-read verified; hash in `literature/claims.jsonl`)
FBA 1999 [3]	lit-009	`3E330BC1...12890B5` (`cpslq.pdf`, 218 997 B)
Bailey 1998 §2 (PSLQ) [2]	lit-010	`5E435CAF...4895` (`node3.html`, 11 344 B); `8BEB376B...6A00` (`paper.html`, 2 123 B)

Canonical artifacts. The two load-bearing JSONL files are `harness/sweep_output/m32a_primary_cascade.jsonl` (M3.2a, 5 649 B) and `harness/sweep_output/m32b_empirical_sweep.jsonl` (M3.2b, 363 709 B); both are frozen at `gold/M3` = `9c2702d`. The aggregate-verdict audit produced by `harness/_audit_m32b_summary.py` on the same data is saved at `harness/sweep_output/m32b_summary.md` (also at `gold/M3`).

Wall-clock. On a single core of an Intel-class 2024 laptop CPU with 16 GB RAM: M3.2a primary cascade ≈ 66.5 min; M3.2b full empirical sweep ≈ 49.6 min (90.5% on the primary n=15 cascade, ~ 4.7 min on the 64 empirical sub-bases); H10 pre-canonical dry-run ≈ 113 s.

B The AEAL heuristic-accumulation methodology

The present work was conducted under an explicit-evidence-acceptance-and-logging (AEAL) discipline: every load-bearing claim is required to carry an explicit evidence class, every halt event is logged, every heuristic is named and dated, and any change to the bounded hypothesis incurs a mutation-log entry. The discipline was iteratively shaped by the work as it proceeded; the present appendix records the heuristic set that crystallized over the mission's six milestones M1–M5 plus the present M6, as a small but real methodological contribution distinct from the K_0 result.

B.1 Heuristic accumulation H1–H11

The eleven heuristics canonical to this work are:

H1. PSLQ cascading-precision triage: Any candidate relation is verified by re-running PSLQ at 2× and 4× precision with the same coefficient bound; transient false positives at low precision are rejected.

H2. Literature-ledger prioritization for the bounded survey: Literature claims that bound the central hypothesis are loaded into the ledger first; tangential citations are not pursued.

H3. Stall-diagnostic timing within wall-clock caps: A scheduled stall-diagnostic at fixed wall-clock checkpoints separates budget-exhaustion from algorithmic stalls.

H4. Capability-gap escalation discipline: If a step of the mission requires machinery not in the declared capability stack, the gap is documented (not papered over) and the mission halts for operator review.

H5. Mutation-log entry discipline: Any change to the bounded hypothesis (basis shape, predicate form, success criterion) incurs a mutation-log entry; field-map updates (citation strength, status promotions within an established taxonomy) do not consume mutation budget.

H6. Namespace audit on independence scoring: Independence claims between two basis classes are audited at the namespace level (content / machinery / attribution).

H7. Functional verification on capability claims: A claimed capability (a tool, an algorithm, a package) is verified by function-call rather than name resolution alone.

H8. Paper-read verification on dependency literature claims: Load-bearing literature claims are verified by direct reading of the primary source (paper, preprint, supplementary material); web aggregators, citation databases, and LLM-mediated summaries are not sufficient.

H9. Theorem-vs-heuristic classification on cited bounds: Cited numerical bounds are reported under a four-class taxonomy (rigorous_theorem, proven_corollary, field_standard_practice, empirical_heuristic), with the class explicit in the manuscript and in the canonical JSONL output.

H10. Full-regime dry-run mandatory before canonical execution: Any sweep introducing a new $(P, n, \text{basis structure})$ regime requires a full-regime dry-run at canonical precision before canonical execution; the dry-run catches code paths that exist but have never been exercised at canonical scale, distinct from H7's "tool isn't installed but name resolves."

H11. Class-of-error exhaustive propagation on bundled fixes: Once a class-of-error pattern is identified at any single site in an AEAL-bound artifact, the fix-site enumeration is exhausted via repository-wide pattern grep (including near-miss variants) before the fix bundle is committed; sibling-class sites are either fixed in the same bundle or catalogued as deliberate exceptions, with an independent-pass review providing a final enumeration check. Distinct from H10's harness-execution audit and from H7's capability-claim audit, H11 catches the partial-fix-propagation failure mode where an initial-site repair leaves the same class-of-error live at unrepaired sibling sites.

Six of these eleven (H6–H11) were installed mid-mission. Five emerged from specific halt events: H6 from a target-shortlist overlap audit; H7 from a capability-probe false-positive on a PARI/GP install state; H8 from a literature-fidelity catch on an AI-aggregated claim later refuted by paper-read; H9 from a heuristic-vs-theorem catch on the $H \approx 10^{P/n}$ scaling formula; H10 from a harness-execution catch when a code path was exercised at canonical precision for the first time. The sixth (H11) was installed proactively at M6 closure as a codification of a class-of-error propagation discipline the mission itself demonstrated during this paper's own layer-3 revision, where the initial-site repair of a displayed empirical-height formula left sibling-site residuals at six further locations until exhaustive grep enumeration plus an independent-pass review cleared them.

B.2 Bidirectional verification: heuristics apply to operator framing as well as agent output

A property of the heuristic set that emerged during the mission is its bidirectional application: H8 fires not only when the agent cites a literature claim, but also when the operator's framing of a claim is under examination. For example, the resolution of one halt event required determining whether a claim that "FBA 1999 states $H \approx 10^{P/n}$ as a theorem" was operator-introduced or agent-introduced; H8 paper-read on [3] established that neither party had cited the claim as a theorem of FBA, and the H9 classification was upgraded accordingly. The bidirectionality is structural: the AEAL discipline is not an agent-fidelity audit; it is a joint operator-and-agent fidelity audit against canonical artifacts.

B.3 Health-score retrospective

We close with four health-score observations across the full mission. The claim-to-sweep ratio (load-bearing claims in the manuscript per candidate sub-basis tested) is approximately 5/65: the manuscript makes few claims, the sweep tests many candidates (sub-basis-level convention; the audit-facing evidence-unit-level convention yields $5/130 \approx 0.038$, per `handoff/agent_health_score_methodology.md` §Axis 1). The reproduce-command coverage is approximately 100%: every numerical claim in the manuscript has a corresponding canonical command in Appendix A. The capability-gap honesty is positive: a fundamental gap is reported as such in Section 4.3 rather than papered over with additional numerics. This figure has additionally been audited against the suspected-but-undocumented-gap honest self-assessment procedure — silent capability verification, silent precision workaround, silent sweep extension, silent subgoal folding, silent scope-cession re-entry — all five questions returning negative across the M1–M6 retrospective (record: `handoff/agent_health_score.md` §Axis 3). The literature-fidelity honesty is positive: two paper-read events ([3] mid-mission; [2] at M6 Step 1) upgraded the manuscript's empirical and rigorous-tier framings against the canonical primary sources, in both cases strengthening the framing without weakening any verification class.

References

[1] David H. Bailey, Jonathan M. Borwein, and Richard E. Crandall. On the Khinchin constant. *Mathematics of Computation*, 66(217):417–431, 1997. §4 Test 1 (pure-power) records null at

D=50, $H \leq 10^{70}$, 7350 dps. §4 close names log K_0, $(\log K_0)(\log 2)$, and "one of many other forms involving K_0" as open. lit-002.

[2] David H. Bailey and Simon Plouffe. Recognizing numerical constants. In *Organic Mathematics: Proceedings of the Workshop held in Burnaby, BC, December 12–14, 1995*, volume 20 of *CMS Conference Proceedings*, pages 73–88. American Mathematical Society, 1998. §2 "The PSLQ Integer Relation Algorithm" is the simplified-PSLQ exposition mpmath cites verbatim in its docstring (URL `http://www.cecm.sfu.ca/organics/papers/bailey/paper/html/node3.html`). Paper-read verified 2026-05-16: SHA-256 `8BEB376B...6A00` (`paper.html`), SHA-256 `5E435CAF...4895` (`node3.html`). §2 "general rule" passage is the primary-source warrant for the field-standard-practice class. lit-010.

[3] Helaman R. P. Ferguson, David H. Bailey, and Steve Arno. Analysis of PSLQ, an integer relation finding algorithm. *Mathematics of Computation*, 68(225):351–369, 1999. Theorem 1 (per-iteration lower bound $M_x \geq 1/\max_i |h_{i,i}|$ on the smallest relation Euclidean norm) and Corollary 2 (termination iteration count) are the rigorous-tier anchor for this manuscript. Theorem 3 (overshoot bound, requires $\gamma > \sqrt{4/3}$ strict) is *not* invoked. Paper-read verified, SHA-256 `3E330BC1...12890B5`. lit-009.

[4] A. Ya. Khinchin. Metrische Kettenbruchprobleme. *Compositio Mathematica*, 1:361–382, 1934.

[5] Papanokechi. khinchin-signature-pslq: a precision-controlled null on the signature statistic family on π's continued fraction. GitHub repository, MIT-licensed, April 2026. Released 2026-04-17; identifies algebraic-combination relations of K_0 with named constants as out-of-scope open extension. lit-001 in this manuscript's literature ledger.